

LESSON 2.4

SQUARE ROOTS AND CUBE ROOTS



LESSON INTRODUCTION

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

LEARNING TARGETS

- I can extend my understanding of square roots and cube roots.
- I can compare square roots and cube roots.
- I can approximate square roots and cube roots.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system.

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

ESSENTIAL QUESTIONS

- How are square roots and cube roots similar and different?
- How do you approximate the value of nonperfect square roots and of cube roots?
- How do you plot square roots and cube roots on a number line?

LESSON NARRATIVE

Students apply their knowledge of square roots to develop an understanding of cube roots. Students will explore how to approximate square and cube roots of nonperfect squares and cubes, recognizing that these are irrational numbers. Students will also locate square and cube roots on a number line.

COMPONENT SUMMARY

2.4.1 WARM-UP (~5 MIN.)

Complete a number pattern

2.4.3 EXPLORATION (10 MIN.)

Estimate the values of nonperfect square roots using the nearest perfect squares

2.4.5 GUIDED PRACTICE (10 MIN.)

Plot square and cube roots on a number line

2.4.2 COLLABORATION (10 MIN.)

Summarize learning related to solving equations and evaluating square and cube roots

2.4.4 EXPLORATION (5-10 MIN.)

Estimate nonperfect cube roots using the nearest perfect cubes

2.4.6 WRAP-UP (~5 MIN.)

Reflect on successes and areas of perseverance during today's lesson

RESOURCES

GLOSSARY

Key Terms:

- Irrational number
- Principal square root

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Consider preparing a classroom number line from zero to seven using chart paper, where students can plot the square roots and cube roots when debriefing 2.4.5 Guided Practice.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse square roots with cube roots because they do not pay attention to the index.

THINGS TO CONSIDER

- Students know the definition of and have worked with rational numbers in 7th grade.
- Students will more formally approximate the value of irrational numbers in subsequent lessons. This lesson is built for students to explore the concepts.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

2.4.2 Collaboration offers an opportunity for students to develop a deeper understanding of how to estimate principal square roots by engaging in rich mathematical conversations with their partners and then seeing and hearing from a variety of their classmates during the whole-class debrief. Before engaging in these discussions, prompt students to first attempt the questions independently.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students explore how to approximate nonperfect square in 2.4.3 Exploration and cube roots in 2.4.4 Exploration by applying their knowledge of perfect squares and cubes. Students begin developing fluency with these skills as practice exercises and discussions with their partners and the whole class, which will continue during subsequent lessons.

DIFFERENTIATE INSTRUCTION

In 2.4.4 Exploration and 2.4.5 Guided Practice, prompt students to generate a list of the perfect cubes to reference as a scaffold, similar to the list of perfect squares they generated in 2.4.3 Exploration. Also, consider providing students with sentence stems to structure their responses in 2.4.6 Wrap-Up.

Additional differentiation opportunities are denoted throughout the lesson guide.

Complete the list of the first 15 perfect squares.

$1^2 = \underline{1}$	$2^2 = \underline{4}$	$3^2 = \underline{9}$
$4^2 = \underline{16}$	$5^2 = \underline{25}$	$6^2 = \underline{36}$
$7^2 = \underline{49}$	$8^2 = \underline{64}$	$9^2 = \underline{81}$
$10^2 = \underline{100}$	$11^2 = \underline{121}$	$12^2 = \underline{144}$
$13^2 = \underline{169}$	$14^2 = \underline{196}$	$15^2 = \underline{225}$

2.4.1 WARM-UP: NUMBER PATTERN

TEACHER MOVES

Discuss how the relationship between the values could be described as subtractive or additive. The subtractive description is provided in the answer key. The additive relationship is that the sum of any two adjacent numbers is the number between them in the next row up.

QUESTIONING

Encourage students to make their own puzzles that involve patterns. Have partners exchange patterns, find missing values, and explain their reasonings. Challenge students to incorporate perfect squares or roots into their puzzles.



2.4.1 Warm-Up: Number Pattern

A pattern is shown with one value missing.

		47		
	25		22	
12		13		9
2		10		3
				6

- Determine the value of the empty cell in the pattern.
47

- Explain how you determined the missing value.
Answers will vary. For a diagonal pair, such as 12 and 10, the difference is 2, the value to the left of the lower number. This pattern holds for the diagram.



2.4.2 Collaboration: Square Roots vs. Cube Roots

Work with your partner to complete the following.

- Damion knows that $8^2 = 64$ and $4^3 = 64$. He says this means the following equations are all true.

Equation A	$\sqrt{64} = 8$
Equation B	$\sqrt[3]{64} = 4$
Equation C	$\sqrt{-64} = -8$
Equation D	$\sqrt[3]{-64} = -4$

- Discuss with your partner whether or not you agree that all of the equations Damion wrote are true.
Answers will vary. We agree that Equations A, B, and D are correct but Equation C is incorrect. This is because you cannot find the square root of a negative number. Equation C is incorrect because the square root of -64 is a non-real solution.
- If Damion wrote any false equation(s), explain what makes them false.
If all of Damion's equations are true, explain how you know.
Answers will vary. Equation C is incorrect because you cannot find the square root of a negative number. Equation C is incorrect because the square root of -64 is a non-real number.

- Complete each statement.

- $\sqrt[3]{-1}$ is ☐ equal to -1 ☐ not a real number because

☐ $-1 \times -1 \times -1 = -1$.
☐ $\quad \times \quad = -1$.
☐ no real number multiplied by itself results in -1 .

- $\sqrt{-1}$ is ☐ equal to $\quad$ ☐ not a real number because

☐ $\quad \times \quad \times \quad = -1$.
☐ $\quad \times \quad = -1$.
☐ no real number multiplied by itself results in -1 .

- Work with your partner to complete the table by choosing a value of k based on the inequality. Then, determine if the result is real or non-real.

Description	Example	Real or Non-Real Result(s)
$\sqrt{k}$ where $k > 0$	<i>Answers will vary.</i> $\sqrt{4} = \pm 2$; $\sqrt{9} = \pm 3$	☐ real ☐ non-real
$\sqrt{k}$ where $k < 0$	<i>Answers will vary.</i> $\sqrt{-25} = \text{non-real}$; $\sqrt{-16} = \text{non-real}$	☐ real ☐ non-real
$\sqrt[3]{k}$ where $k > 0$	<i>Answers will vary.</i> $\sqrt[3]{64} = 4$, $\sqrt[3]{8} = 2$	☐ real ☐ non-real
$\sqrt[3]{k}$ where $k < 0$	<i>Answers will vary.</i> $\sqrt[3]{-64} = -4$ $\sqrt[3]{-8} = -2$	☐ real ☐ non-real

- Write a short explanation of when a non-real result will occur.
Answers will vary. If you multiply two positive numbers the answer is positive, and when you multiply two negative numbers, the answer is also positive. Therefore, a non-real result will occur when you try to find the square root of a negative number because the result of squaring a real number is always positive.

2.4.2 COLLABORATION: SQUARE ROOTS VS. CUBE ROOTS

TEACHER MOVES

Some students may also identify Equation A as an incorrect answer because only one of the two solutions is listed. Consider using this as an opportunity to discuss why perfect squares have two roots but perfect cubes only have one. Relate this to the fact that negative radicands are possible for cube roots. Questions 2 illustrates this by having students expand out the multiplication.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

DIFFERENTIATE INSTRUCTION

Students may struggle to interpret the algebraic description in the table in question 3. Consider providing a written or verbal description and a few possible values students can use as examples. Modeling with a Think Aloud may help students understand the task expectations. Also, remind students that if no index is written, it is assumed to be 2.

2.4.3 EXPLORATION: SQUARE ROOTS OF NON-PERFECT SQUARES

TEACHER MOVES

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group in 2.4.5 Guided Practice. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

QUESTIONING

Ask students if they notice any patterns in consecutive square numbers. Have them explain the patterns using words, models, or a combination of both.

Side Length: 1 2 3 4 5 6 7...

Perfect squares: 1 4 9 16 25 36 49...

Differences: 3 5 7 9 11 13...

Prompt students to look back at the geometric representations of perfect squares to explain the pattern.



2.4.3 Exploration: Square Roots of Non-Perfect Squares

There are an infinite number of perfect squares in the real number system.

- Complete the list of the first 15 perfect squares.

$$1^2 = \underline{1} \qquad 2^2 = \underline{4} \qquad 3^2 = \underline{9}$$

$$4^2 = \underline{16} \qquad 5^2 = \underline{25} \qquad 6^2 = \underline{36}$$

$$7^2 = \underline{49} \qquad 8^2 = \underline{64} \qquad 9^2 = \underline{81}$$

$$10^2 = \underline{100} \qquad 11^2 = \underline{121} \qquad 12^2 = \underline{144}$$

$$13^2 = \underline{169} \qquad 14^2 = \underline{196} \qquad 15^2 = \underline{225}$$

- Check your answers with your partner.

The square roots of perfect squares are rational numbers because they can be expressed as a ratio of two integers.

Square roots of non-perfect squares, such as $\sqrt{14}$, are **irrational numbers**.



Irrational numbers are real numbers that cannot be expressed as a ratio of two integers.

The decimal expansions of irrational numbers do not terminate nor repeat. Therefore, the values of **principal square root** of a non-perfect square can only be approximated.



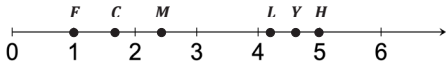
The **principal square root** is the positive square root of a positive real number.

Perfect squares can be used to estimate the value of the principal square root of non-perfect squares.

3. Work with your partner to complete the following to approximate the principal square root of $\sqrt{14}$.
- a. Complete the inequality to show which two perfect squares 14 lies between.
 $9 < 14 < 16$
 - b. Use your answers to Part A to indicate which two square roots $\sqrt{14}$ lies between.
 $3 < \sqrt{14} < 4$
 - c. Discuss with your partner which square root the value of $\sqrt{14}$ is closer to.
 4
 - d. Plot $\sqrt{14}$ on the number line to indicate its approximate value based on your discussion.



The approximate locations of six square roots are plotted on the number line shown.



4. Work with your partner to determine which point matches the approximate value of each square root given in the table.

Square Root	Point
$\sqrt{3}$	C
$\sqrt{21}$	Y
$\sqrt{1}$	F
$\sqrt{6}$	M
$\sqrt{18}$	L
$\sqrt{25}$	H

2.4.3 EXPLORATION: SQUARE ROOTS OF NON-PERFECT SQUARES (CONTINUED)

ENRICHMENT

If students are stuck as they work with their partners through question 3, consider asking the following:

- How did you determine which two perfect squares 14 lies between?
- What is the difference between 9 and 14? Between 14 and 16? How can this help you to determine which square root $\sqrt{14}$ is closest to?
- How will you show this on the number line?

TEACHER MOVES

Question 3 is meant to preview the thinking used to approximate square roots. They will go more formally into finding approximate values in the 2.4.5 Guided Practice later in this lesson as well as in Lesson 5.

TEACHER MOVES

Listen as students are collaborating to hear how they are reasoning about the values, in particular L and Y. There are two square roots between 4 and 5, but it's clear which is which using reasoning similar to the previous question. Listening to students' reasoning will help the teacher to determine what to focus on in the first question in the 2.4.5 Guided Practice.

2.4.3 EXPLORATION: SQUARE ROOTS OF NON-PERFECT SQUARES (CONTINUED)

TEACHER MOVES

There are four different ways to correctly complete this statement. Their choices in the first box are all the square roots of nonperfect squares. Students can choose any of those four choices, and then the remaining correct choices will change based on that decision.

If time allows, teachers could have students complete this part independently and then compare with their partner so they could potentially discuss two different correct ways to complete the statement or the teacher could have them work together on this. Observational data about students' comfort with these concepts should be used to make this decision.

TEACHER MOVES

Think-Pair-Share

Offer students an opportunity to develop a deeper understanding of how to estimate principal square roots by engaging in rich mathematical conversations with their partners and then seeing and hearing from a variety of their classmate during the whole-class debrief. Before engaging in these discussions, prompt students to first attempt the questions independently.

5. Complete the statement to explain how you determined the approximate location of the point selected.

Answers will vary based on response. There are four different ways to correctly complete the statement.

Point

☐ C
☐ M
☐ L
☐ Y

 is between

☐ 1
☐ 2
☐ 3
☐ 4
☐ 5

 and

☐ 1,
☐ 2,
☐ 3,
☐ 4,
☐ 5,

 but

closer to

☐ 1
☐ 2
☐ 3
☐ 4
☐ 5

 because

☐ $\sqrt{3}$
☐ $\sqrt{21}$
☐ $\sqrt{6}$
☐ $\sqrt{18}$

 is between $\sqrt{\quad}$

and $\sqrt{\quad}$, but closer to $\sqrt{\quad}$.

Point C is between 1 and 2, but closer to 2 because $\sqrt{3}$ is between $\sqrt{1}$ and $\sqrt{4}$ but closer to the $\sqrt{4}$.

Point M is between 2 and 3, but closer to 2 because $\sqrt{6}$ is between $\sqrt{4}$ and $\sqrt{9}$ but closer to the $\sqrt{4}$.

Point L is between 4 and 5, but closer to 4 because $\sqrt{18}$ is between $\sqrt{16}$ and $\sqrt{25}$ but closer to the $\sqrt{16}$.

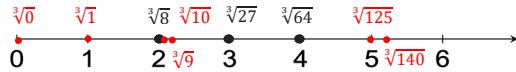
Point Y is between 4 and 5, but closer to 5 because $\sqrt{21}$ is between $\sqrt{16}$ and $\sqrt{25}$ but closer to the $\sqrt{25}$.



2.4.4 Exploration: Estimating Cube Roots

Similar to the square roots of non-perfect squares, the cube roots of some numbers are irrational. The values of such cube roots can be approximated using the cube roots of perfect cubes.

1. A number line is shown with values indicated at specific points.



- a. Explain why $\sqrt[3]{8}$ is placed in the same location as 2 on the number line.
Answers will vary. It is placed there because 2 is the cubed root of 8. You multiply $2 \times 2 \times 2$ to get 8 therefore the cubed root of 8 is 2.
- b. Label the locations of the following cube roots on the number line.
 $\sqrt[3]{1}$, $\sqrt[3]{125}$, and $\sqrt[3]{0}$
- c. Plot and label the approximate locations of the following cube roots on the number line.
 $\sqrt[3]{9}$, $\sqrt[3]{10}$, and $\sqrt[3]{140}$
- d. Compare your answers to your partner's. Discuss any differences and make revisions to your work as needed.

2.4.4 EXPLORATION: ESTIMATING CUBE ROOTS

DIFFERENTIATE INSTRUCTION

Consider prompting students to generate a list of the perfect cubes from one to six to reference.

ENRICHMENT

If students are stuck as they work with their partners through question 1, consider asking the following:

- How did you determine which two perfect cubes 9 lies between?
- What is the difference between 9 and 8? Between 9 and 27?
- How can this help you to determine which cube root $\sqrt[3]{9}$ is closest to?
- How can this be shown on the number line?

2.4.5 GUIDED PRACTICE: LOCATING SQUARE ROOTS AND CUBE ROOTS ON A NUMBER LINE

DIFFERENTIATE INSTRUCTION

Use observational data from the 2.4.3 and 2.4.4 Explorations to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in the Explorations.

QUESTIONING

Challenge students to locate and plot other possible square roots of whole numbers on the number line.

- How many numbers do you think you will plot on the number line? (Fourteen.)
- How did you determine this? (Six is square root of 36, and 7 is square root of 49. There are 14 numbers between 36 and 49, inclusive.)
- Select one of the square roots and explain how you determined where to plot it on the number line.

TEACHER MOVES

k and q will be approximate locations. q should be slightly to the right of 3, and k should be slightly to the left of 3. It is not the expectation that students go through the steps similar to the previous question to find a more precise estimation, rather that they can reason about the approximate locations using what they know about perfect squares and cubes.

Students may be confused about why q is to the right and k is to the left of 3. Remind them to check the index to note that although 24 is larger than 10, k is equal to the cube root of 24, while q is equal to the square root of 10.

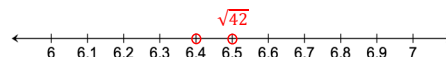


2.4.5 Guided Practice: Locating Square Roots and Cube Roots on a Number Line

1. Consider $\sqrt{42}$.

- Between which two perfect squares does 42 lie?
36 and 49
- Between which two integers is the value of $\sqrt{42}$?
6 and 7
- Corey said, "42 is about halfway between 36 and 49, but slightly closer to 36, so I think a good estimate for $\sqrt{42}$ would be about halfway between 6 and 7, but slightly closer to 6."

Circle two of the values shown on the number line that Corey could use to check his thinking.



- Square both values circled in Part C to determine which is a better approximation for $\sqrt{42}$.
 *$(6.4)^2 = 40.96$ and $(6.5)^2 = 42.25$
 42.25 is closer to 42 than 40.96 so 6.5 is a better approximation of the square root of 42.*
- Plot and label the approximate location of $\sqrt{42}$ on the number line above based on your work in Part D.

2. Six equations are shown.

$$a = \sqrt{9}$$

$$b^3 = 8$$

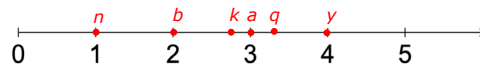
$$\sqrt{10} = q$$

$$1 = n^3$$

$$k = \sqrt[3]{24}$$

$$\sqrt[3]{64} = y$$

- Plot and label the approximate locations of $a, b, q, n, k,$ and y on the number line.



- Choose one value and explain how you determined its location on the number line.
Answers will vary. I chose to put the $\sqrt{10}$ between 3 and 4 on the numberline but closer to 3 because $3^2 = 9$ and $4^2 = 16$. I put the $\sqrt[3]{64}$ at 4 because $4 \times 4 \times 4 = 64$.

**2.4.6 Wrap-Up: Growth Mindset**

"Don't be afraid of failure. This is the way to succeed."
- LeBron James

1. Complete each statement.

- a. Today I found a way to succeed at...

Answers will vary. Today I found a way to succeed at estimating the value of square and cube roots. Today I found a way to succeed at graphing the location of squares and cube roots on a number line.

- b. Today I did not give up on learning...

Answers will vary. Today I did not give up on learning how to estimate the value of square and cube roots that are not from perfect squares. Today I did not give up on learning how to graph the location of squares and cube roots on a number line.

2.4.6 WRAP-UP: GROWTH MINDSET**DIFFERENTIATE INSTRUCTION**

Consider providing sentence stems, such as:

- "I was successful in today's work when I _____."
- "It was challenging to _____, but I did not give up, because _____."